L298N Dual Full-Bridge Driver

STMicroelectronics Specification | Page 1

1. Device Overview & Maximum Ratings

The L298N is an integrated monolithic circuit in a 15-lead Multiwatt package. It is a high voltage, high current dual full-bridge driver designed to accept standard TTL logic levels and drive inductive loads such as relays, solenoids, DC motors, and stepper motors.

Parameter	Rating Value	Operating Condition
Motor Supply Voltage (Vs)	Up to 46.0V DC	Minimum Vs = 4.5V; typical 12V to 24V
Logic Supply Voltage (Vss)	4.5V to 7.0V DC	Nominal 5.0V for internal logic gates
Peak Output Current (per channel)	3.0A (non-repetitive)	Pulse duration t <= 100 us
Continuous Output Current	2.0A per channel	Total continuous 4A with proper heatsink
Input Logic High (VIH)	2.3V to Vss	Compatible with 3.3V and 5V microcontrollers

2. Pinout & H-Bridge Control Logic (Multiwatt 15)

• Pin 4 (Vs): Motor power supply (+12V to +35V DC).

• Pin 9 (Vss): Logic power supply (+5V DC).

• Pin 8 (GND): Common circuit ground. Must be connected to microcontroller ground!

• Pin 5 (IN1) & Pin 7 (IN2): Bridge A direction inputs. IN1=HIGH, IN2=LOW drives Forward; IN1=LOW, IN2=HIGH drives Reverse; IN1=IN2 stops motor.

• Pin 6 (ENA): Bridge A Enable input. Apply PWM signal (0-100% duty cycle) to regulate motor speed.

• Pin 2 (OUT1) & Pin 3 (OUT2): Bridge A outputs connected to Motor A terminals.

• Pin 10 (IN3), Pin 12 (IN4), Pin 11 (ENB), Pin 13 (OUT3), Pin 14 (OUT4): Bridge B controls for Motor B.

• Pin 1 & Pin 15 (Current Sensing A & B): Connect to GND directly, or via 0.5 Ohm sense resistors for current monitoring.

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3. Inductive Spike Protection (Flyback Diodes)

CRITICAL PROTECTION REQUIREMENT: DC motors generate extreme inductive back-EMF voltage spikes (exceeding 100V) when commutating or stopping. The internal bipolar transistors of the L298N do NOT have internal suppression diodes.

• Each output pin (OUT1, OUT2, OUT3, OUT4) MUST be clamped to Vs and GND using 8 external fast-recovery diodes.

• Recommended Diode: 1N5819 Schottky diode or 1N4937 fast recovery diode (trr < 200 ns). Do NOT use standard 1N4007 rectifier diodes as their slow reverse recovery time will damage the L298N.

4. Troubleshooting & Thermal Diagnostics

• Issue 1: Motor driver smokes or heats up to > 100°C within seconds. Cause: Bipolar transistor saturation loss. The L298N has an internal Vce saturation drop of ~2.0V to 3.0V. At 2A load, internal power dissipation is P = 2.5V * 2A = 5 Watts! Solution: A large aluminum heatsink with thermal paste is mandatory for currents above 0.8A.

• Issue 2: Motor does not turn, but logic LEDs light up. Cause: Vs (Pin 4) disconnected or Enable jumper missing. If ENA (Pin 6) is left floating, Bridge A is disabled. Solution: Tie ENA to Vss or apply PWM.

• Issue 3: Microcontroller crashes or reboots when motor reverses direction. Cause: Ground loop noise or motor inductive back-EMF resetting the MCU. Solution: Separate motor power supply from logic supply and connect grounds at a single star-ground point.